



Jacobi Orthogonal Polynomials in ANN

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Related work

- Chu Kwong Chak "Orthogonal polynomials neural network for function approximation and system modeling" [2]
 - Explains the pros for using orthogonal polynomials in NN. reduces the need of complexity through iterations
 - Orthogonal basis allows much smaller magnitude of parameter
 - Different model used
- Shen, Jie "Orthogonal polynomials and related approximation results." *Spectral Methods: Algorithms, Analysis and Applications* (2011): 70-80 [7].
 - Proofs and alternative equations for the Jacobi orthogonal polynomials and its derivative.
- H. Zhang "Risk Assessment of Substation based on Hermite orthogonal basis Feedforward Neural Network data Fusion algorithm" [3]
 - Used a paralyzation technique avoid using Back propagation that works with orthogonal polynomials.



Orthogonal Polynomials

- Set of functions that follow an orthogonal relation from each other
 - Each set of functions map out a different space or dimension from each other.
- Similar to vectors, we can define each value to a polynomial function
 - $\langle f(x_0), f(x_1), \dots, f(x_n) \rangle * \langle g(x_0), g(x_1), \dots, g(x_n) \rangle = 0$
 - $\sum [x; x \in X] f(x)g(x) = 0$



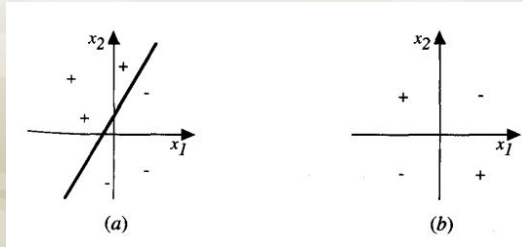
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 - $\sum [x; x \in X] f(x)g(x) = 0$
- Goes into the general equation
 - $\int f(x)g(x)dx = 0$
- Including a “weight” function
 - $\int f(x)g(x)w(x)dx = 0$



Why Orthogonal Polynomials?

- Polynomials can solve for nonlinear solutions
- There are many different polynomial functions
- Orthogonality provides a set of polynomial functions to use
- More efficient because each polynomial will be guaranteed to be independent of each other.
- Allows Neural Networks to search through the hypothesis space faster.





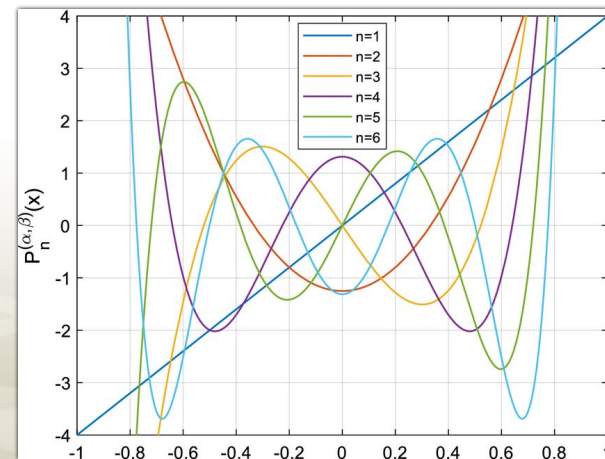
Jacobi Polynomials

- Jacobi Polynomials are a set of orthogonal polynomials that are derived from $w(x) = (1-x)^\alpha (1+x)^\beta$ on the range of $[-1, 1]$
 - Where α and β are constants that are > -1
- Recurrence formula

$$J_{n+1}^{\alpha,\beta}(x) = (a_n^{\alpha,\beta} - b_n^{\alpha,\beta})J_n^{\alpha,\beta}(x) - c_{n-1}^{\alpha,\beta}J_{n-1}^{\alpha,\beta}(x), \quad n \geq 1$$
$$J_0^{\alpha,\beta}(x) = 1, \quad J_1^{\alpha,\beta}(x) = \frac{1}{2}(\alpha + \beta + 2)x + \frac{1}{2}(\alpha - \beta)$$

○ where

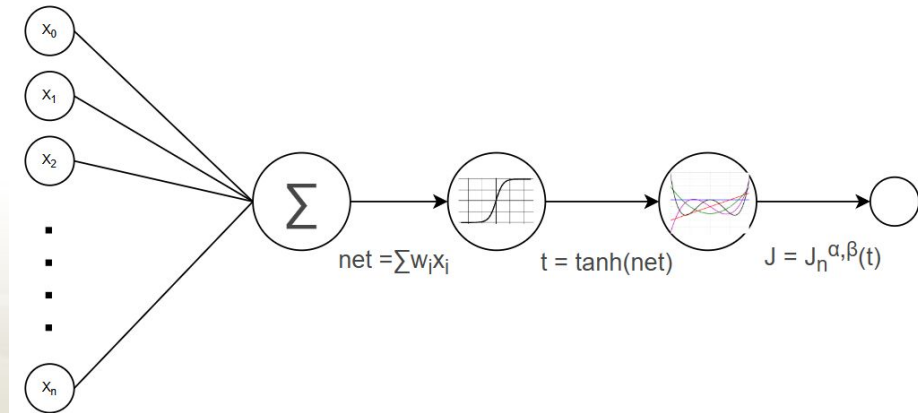
$$a_n^{\alpha,\beta} = \frac{(2n+\alpha+\beta+1)(2n+\alpha+\beta+2)}{2(n+1)(n+\alpha+\beta+1)}$$
$$b_n^{\alpha,\beta} = \frac{(\beta^2 - \alpha^2)(2n+\alpha+\beta+1)}{2(n+1)(n+\alpha+\beta+1)(2n+\alpha+\beta)}$$
$$c_n^{\alpha,\beta} = \frac{(n+\alpha)(n+\beta)(2n+\alpha+\beta+2)}{(n+1)(n+\alpha+\beta+1)(2n+\alpha+\beta)}$$





Implementation

- $\tanh(x)$ is used as a second activation function to make sure the net function stays between $(-1,1)$.
- Alpha and Beta values are set as “magic numbers”.
 - Pre calculated before training.





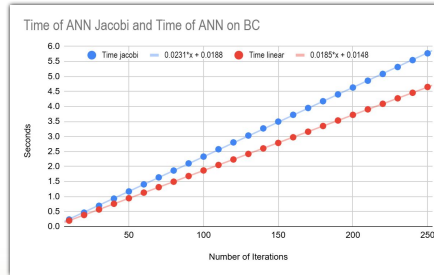
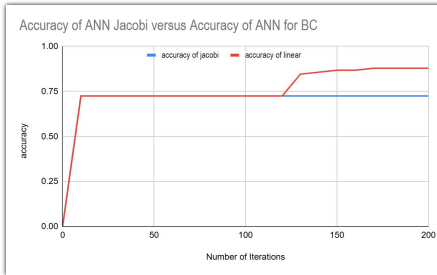
Databases

- Breast Cancer Wisconsin (Diagnostic) [5]
 - Target is on patients with or without breast Cancer
 - Attributes are the shapes of 3 cell nuclei found in the patient
- Default Credit Card Client [4]
 - Research on individuals in taiwan and their default pay as risk management
 - Target value is if the individual will have default pay next month
 - Attributes are the individual characteristics (age, sex, education, etc.) and bill payments in the last couple of months



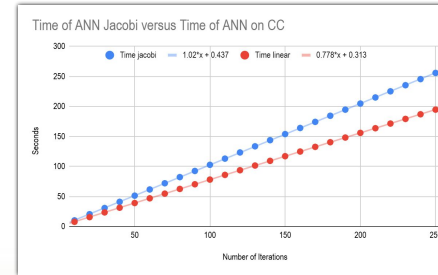
Results

Breast Cancer



- Standard ANN was able to pass the local minimum problem with enough iterations.
- Jacobi polynomials could not pass this.
- Changes to alpha or beta did not change outcomes of the dataset
- Slope ratio: 1.25

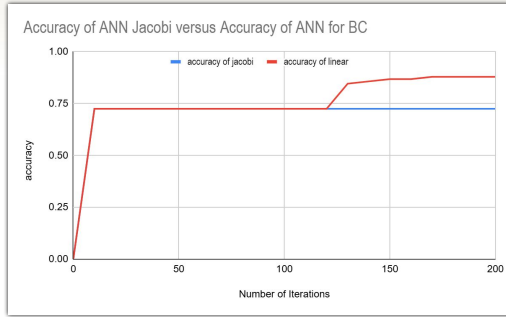
Credit Card



- Stayed consistent 0.774% accuracy for both models
- Slope ratio: 1.31



Results



- NN using Jacobi has problems getting stuck on local minimums
- Jacobi training time runs close to linearly for changes in number of hidden



Results

ANN using Jacobi Polynomials

training time	0.0252	0.0246	0.026	0.0252	0.0252	0.0255	0.0286	0.0251	0.025	0.0246
# of iterations	1	1	1	1	1	1	1	1	1	1

average	std Dev
0.0255	0.001162373052
1	0

Standard ANN

training time	0.2098	0.0207	0.0411	0.0209	0.0204	0.0838	0.021	0.409	2.335	0.0207
# of iterations	11	1	2	1	1	4	1	22	128	1

average	std Dev
0.31824	0.7196926554
17.2	39.51877191

- Jacobi Polynomials were consistent at finding a high accuracy level quickly
- Average time to reach high values are faster with Jacobi



Conclusion

- Overall time to reach high accuracy values are faster with Jacobi NN
- Jacobi NN is more difficulty passing through the local minimum problem
- Jacobi NN requires 2 more magic numbers (Alpha and Bets) to build it's
 - Cannot dynamically change during learning process (possibly something to work on in the future)



sources

- H. S. Chakraborty, T. Miezaki, M. Oura, and Y. Tanaka, 'Jacobi polynomials and design theory I', *Discret. Math.*, vol. 346, p. 113339, 2022.
DOI: 10.48550/arXiv.2207.10911
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- Image1: <https://www.cs.cmu.edu/~tom/mlbook.html>
- Image2: https://www.researchgate.net/figure/The-graphs-of-the-first-six-Jacobi-polynomials-of-degree-n1-2-6-with_fig1_345895072



Questions?

